

# Day 3 Cheatsheet

## Subsetting Data in R

### Functions

| Library/Package | Piece of code                     | Example of usage                                        | What it does                                                                                                                                                                  |
|-----------------|-----------------------------------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Base R          | <code>nrow(x); ncol(x)</code>     | <code>nrow(x); ncol(x)</code>                           | Get the number of rows and the number of columns in an object <code>x</code> , respectively.                                                                                  |
| Base R          | <code>dim(x)</code>               | <code>dim(x)</code>                                     | Get the number of rows <i>and</i> number of columns in an object <code>x</code>                                                                                               |
| dplyr           | <code>glimpse(x)</code>           | <code>glimpse(mtcars)</code>                            | Get an overview of data frame <code>x</code>                                                                                                                                  |
| dplyr           | <code>slice_sample(x)</code>      | <code>slice_sample(mtcars)</code>                       | See a random subset of the rows of <code>x</code>                                                                                                                             |
| Base R          | <code>data.frame()</code>         | <code>df &lt;- data.frame(1:3)</code>                   | Creates a data frame where the named arguments will be the same length.                                                                                                       |
| Base R          | <code>tibble()</code>             | <code>tibble(mtcars)</code>                             | Creates a tibble from a <code>data.frame</code> or matrix.                                                                                                                    |
| tibble          | <code>rownames_to_column()</code> | <code>df &lt;- column_to_rownames(df, 'new_col')</code> | Transforms the rownames of a data frame into a column (which is added to the start of the data frame). The string supplied as an argument will be the name of the new column. |
| dplyr           | <code>rename()</code>             | <code>df &lt;- rename(df, MPG = mpg)</code>             | Renames designated columns while keeping all variables of the <code>data.frame</code>                                                                                         |
| dplyr           | <code>pull()</code>               | <code>pull(df, 'existing_variable_name')</code>         | Extract a column as a vector                                                                                                                                                  |
| dplyr           | <code>select()</code>             | <code>select(df, 'existing_variable_name')</code>       | Selects columns that match the specified argument                                                                                                                             |
| dplyr           | <code>filter()</code>             | <code>filter(df, mpg &gt; 20)</code>                    | Returns a subset of rows matching the conditions of the specified logical argument                                                                                            |

| Library/Package | Piece of code                     | Example of usage                                                                                                | What it does                                                                                                                                                                                                         |
|-----------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Base R          | <code>==, &lt;=, &gt;=, !=</code> | <code>filter(df, mpg &gt; 20)</code>                                                                            | These are binary operators which allow for the comparison of values in an object. They are handy for use with <code>filter()</code>                                                                                  |
| Base R          | <code>%in%</code>                 | <code>filter(df, mpg %in% c(20,21,22))</code>                                                                   | Checks if the given value(s) on the left side of the operator are in the vector or other R object defined on the right side of the operator. It returns a logical <code>TRUE</code> or <code>FALSE</code> statement. |
| Base R          | <code> &gt;</code>                | <code>df &lt;- df  &gt;</code>                                                                                  | Funnels a data.frame through additional operations                                                                                                                                                                   |
| dplyr           | <code>%&gt;%</code>               | <code>select('variable_name')</code><br><code>df &lt;- df %&gt;%</code><br><code>select('variable_name')</code> | Funnels a data.frame through tidyverse operations                                                                                                                                                                    |
| dplyr           | <code>mutate()</code>             | <code>df &lt;- mutate(df, newcol = wt/2.2)</code>                                                               | Adds a new column that is a function of existing columns                                                                                                                                                             |
| dplyr           | <code>relocate()</code>           | <code>df_carb &lt;- relocate(.data = df, wt, .before = mpg)</code>                                              | Reorder columns in a data frame or tibble                                                                                                                                                                            |
| dplyr           | <code>arrange()</code>            | <code>df &lt;- arrange(df, mpg)</code>                                                                          | Reorders rows in ascending order. <code>arrange(desc())</code> would reorder rows in descending order.                                                                                                               |
| dplyr           | <code>case_when()</code>          | <code>df &lt;- arrange(df, mpg)</code>                                                                          | This function allows you to vectorize multiple <code>if_else()</code> statements. If no cases match, <code>NA</code> is returned.                                                                                    |
| Base R          | <code>colnames()</code>           | <code>colnames(df)</code>                                                                                       | Gets or sets the column names of a matrix or data frame.                                                                                                                                                             |

- See `tidyselect` helpers for handy things to use with `select()`.

\* This format was adapted from the cheatsheet format from AlexsLemonade.